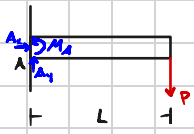


$P = 300 \text{ N}$ $\delta_{\max} = 0.3 \text{ mm}$ $SF = 3$ Material: PLA $S_y = 50 \text{ MPa}$ $E = 3500 \text{ MPa}$ $L \approx 50 \text{ mm}$ $b = 42 \text{ mm}$
 $I = \frac{PL^3}{3E\delta}$

Feature 1



$$\sum F_y = 0$$

$$A_y = 300 \text{ N}$$

$$\sum M_A = 0$$

$$M_A = 300 \cdot 50 = 15000 \text{ N}\cdot\text{mm}$$

$$I = \frac{300 \cdot 50^3}{3 \cdot 3500 \cdot 0.3} = 11904.76 \text{ mm}^4$$

$$I = \frac{b t^3}{12} \rightarrow \frac{42 \cdot t^3}{12} = 11904.76$$

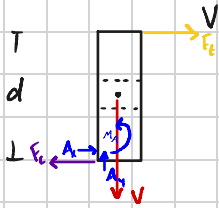
$$t = 15.04 \text{ mm} \rightarrow 16 \text{ mm}$$

$$\sigma_{\text{allow}} = \frac{50}{3} = 16.67 \text{ MPa}$$

$$\sigma = \frac{15000(17)}{73280} = 3.48 \text{ MPa}$$

$$3.48 \leq 16.67 \checkmark$$

Feature 2



$$V = 300 \text{ N} \quad M_A = 300 \cdot 50 = 15000 \quad d = 35 \text{ mm}$$

$$F_e = \frac{M}{d} = \frac{15000}{35} = 428.5714286 \text{ N}$$